

Cosmic Universalism CU-Time

Expanded Outputs and Method Implementation Manual

Full Cosmic Breath Coordinate, Observable-Universe Reference Outputs, and Controlled Astro Integration

A controlled implementation specification for preserving the existing bidirectional Gregorian UTC and CU-Time converter while automatically exposing three coordinated observable-universe reference outputs, documenting their arithmetic, and keeping CU-Time governed by the Cosmic Universalism Statement rather than by empirical cosmology.

Repository	CosmicUniversalismStatement
Protected branch	main
Stable website branch	website
Current website commit	e1c82bf
Current integration tag	website-v1.1-cosmic-breath
Proposed feature branch	feature/cu-time-expanded-outputs
Target route	/CosmicUniversalismStatement/cu-time/
Reference implementation	cosmic_breath_time_converter_v3_0_1.html
Document version	1.1
Verified	July 23, 2026

William Maddock

<https://github.com/willmaddock/CosmicUniversalismStatement>

Document Control

Document	Cosmic Universalism CU-Time: Expanded Outputs and Method Implementation Manual
Version	1.1
Classification	Owner-approved technical specification and controlled implementation manual
Repository	CosmicUniversalismStatement
Protected branch	main
Stable integration branch	website
Current website baseline	e1c82bff1d37732f52b03fa4734cf7e4120b71e9
Current integration tag	website-v1.1-cosmic-breath at 7ac87167c51aad9ae7a31e17d1b2127fc069e15c
Prior CU-Time release	website-v1.1-cu-time-release at 3fa4a37
Proposed feature branch	feature/cu-time-expanded-outputs; not yet created
Target page	website/src/pages/cu-time.astro
Target component	website/src/components/CUTimeConverter.astro
Target route	/CosmicUniversalismStatement/cu-time/
Reference implementation	cosmic_converter/v3_0_0/cosmic_breath_time_converter_v3_0_1.html
Technical baseline inspected	July 23, 2026

Purpose

This manual controls the next CU-Time website phase. The existing converter continues to support Gregorian UTC to CU-Time and CU-Time to Gregorian UTC. The approved enhancement automatically exposes the already-derived observable-universe-aligned coordinate and Big Bang reference chronology, adds one ratio calculation, and expands the page's educational method without creating separate calculators or linking to external converter pages.

Authority and Source Priority

The governing source order for this feature is:

1. William Maddock's owner decisions recorded in this manual.
2. The Cosmic Universalism Statement and the owner-approved CU-Time interpretation of the full Cosmic Breath.
3. Live repository state and the current TypeScript converter engine.
4. The current Website Master Implementation Manual v1.1.
5. The unchanged standalone converter HTML as a behavioral and provenance reference.
6. Historical research files as interpretive or open-research material; they do not silently override owner decisions or production tests.

Safety Rule

This manual authorizes specification and later bounded implementation only. It does not authorize a branch change, source edit, dependency change, commit, push, merge, tag, release, or deployment until that

action is separately approved. The reference HTML remains unchanged.

Contents

Document Control	1
START HERE - Governing Feature Decision	5
1 Part 1 - Governance, Baseline, and Protected Scope	6
1.1 Verified repository baseline	6
1.2 Protected branches and action boundaries	6
1.3 Likely authorized implementation paths	6
1.4 Protected implementation paths	7
2 Part 2 - CU Authority, Statement, and Chronology	7
2.1 The governing CU Statement	7
2.2 CU-Time is the full Cosmic Breath coordinate	7
2.3 No zero as a CU state or origin	8
2.4 Why the Observable Universe appears between B-TOM and C-TOM	8
3 Part 3 - Current Implementation Inspection	9
3.1 Current public route	9
3.2 Current bidirectional interface	9
3.3 Current engine already computes two requested outputs	9
3.4 Only the ratio is a new derived quantity	10
4 Part 4 - Exact Mathematical Specification	10
4.1 Approved exact literals	10
4.2 Gregorian UTC to aligned coordinate	11
4.3 CU-Time to Gregorian UTC	11
4.4 Big Bang reference coordinate	11
4.5 Ratio calculation	11
4.6 Computational precision and empirical uncertainty	12
5 Part 5 - Boundary Semantics and Public Output States	12
5.1 Three age-reference states	12
5.2 Approved boundary wording	12
5.3 No conversion failure at the boundary	13
6 Part 6 - Public Terminology and Result Architecture	13
6.1 Primary output: CU-Time - Full Cosmic Breath Coordinate	13
6.2 Observable-Universe-Aligned CU Coordinate	13
6.3 Derived Age Since the Big Bang	13
6.4 CU Coordinate-to-Observable-Age Ratio	14
6.5 Historical terminology disclosure	14
7 Part 7 - Interface and Content Design	14
7.1 One converter, automatic secondary results	14
7.2 Result-card presentation	15
7.3 Copy behavior	15
7.4 Expanded educational sections below the converter	15

8	Part 8 - TypeScript Data and Formatting Design	16
8.1	Recommended internal model	16
8.2	Recommended public display contract	16
8.3	Formatting policy	17
9	Part 9 - Accessibility, Responsive Design, and Interaction	17
9.1	Accessibility requirements	17
9.2	Responsive requirements	18
9.3	Progressive disclosure	18
10	Part 10 - Test Plan and Acceptance Criteria	18
10.1	Core engine tests	18
10.2	UI adapter tests	18
10.3	Component regression tests	19
10.4	Reference vectors	19
10.5	Full acceptance gate	20
11	Part 11 - Controlled Implementation Phases	20
11.1	Phase 1 - Pure result semantics	20
11.2	Phase 2 - Public display adapter	20
11.3	Phase 3 - Converter result cards	21
11.4	Phase 4 - Route education	21
11.5	Phase 5 - Full acceptance	21
12	Part 12 - Copy-and-Paste Implementation Prompts	21
12.1	Feature-branch preflight prompt	21
12.2	Phase-one implementation prompt	22
12.3	Final review prompt	23
13	Part 13 - Commit, Merge, and Deployment Boundaries	23
13.1	Commit boundary	23
13.2	Merge boundary	23
13.3	Tag boundary	23
14	Appendix A - Source Authority Record	24
15	Appendix B - Owner Decision Record	24
16	Appendix C - Final Acceptance Checklist	25
17	Appendix D - Manual Revision Record	25

START HERE - Governing Feature Decision

Successful Result

The approved product decision is one bidirectional converter, one conversion action, and four coordinated result views. The three new values are calculated automatically from the same input. They are not independent calculators.

Approved result behavior

For **Gregorian UTC to CU-Time**, display:

1. CU-Time - Full Cosmic Breath Coordinate;
2. Observable-Universe-Aligned CU Coordinate;
3. Derived Age Since the Big Bang, or boundary-relative wording;
4. CU Coordinate-to-Observable-Age Ratio when an elapsed age exists.

For **CU-Time to Gregorian UTC**, display:

1. Corresponding Gregorian UTC;
2. Observable-Universe-Aligned CU Coordinate;
3. Derived Age Since the Big Bang, or boundary-relative wording;
4. CU Coordinate-to-Observable-Age Ratio when an elapsed age exists.

Approved exclusions

- no links to the other historical converter pages;
- no separate NASA, Big Bang age, or ratio input forms;
- no claim that NASA created, endorses, or operates CU-Time;
- no claim that the Big Bang begins CU-Time or the Cosmic Breath;
- no TOM-state classification generated from a date input;
- no use of TOM notation as the arithmetic source of the converter;
- no presentation of internal arithmetic zero as a CU state or origin.

Checkpoint

The page enhancement is complete only when both conversion directions expose the same secondary reference values from the same engine result and when the public copy preserves CU authority over empirical comparison.

1 Part 1 - Governance, Baseline, and Protected Scope

1.1 Verified repository baseline

The read-only inspection established:

Item	Verified state
Repository root	/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement
Origin	https://github.com/willmaddock/CosmicUniversalismStatement.git
Current branch	website
Current local and remote HEAD	e1c82bff1d37732f52b03fa4734cf7e4120b71e9
Current status	Clean and synchronized: website...origin/website
Current integration tag	website-v1.1-cosmic-breath remains on 7ac87167c51aad9ae7a31e17d1b2127fc069e15c
Application verification inherited	251 tests across 30 files and an eight-route Astro build at the preceding application integration checkpoint
Documentation-only commit	e1c82bf; added manuals without changing application source

This manual did not rerun the application tests. The 251-test record is the most recent verified application gate inherited from the Cosmic Breath integration. A future CU-Time implementation must run the current full suite and report its new count rather than assuming 251 remains exact.

1.2 Protected branches and action boundaries

- `main` remains protected.
- `website` is the stable integration branch.
- The proposed implementation branch is `feature/cu-time-expanded-outputs`; it does not yet exist.
- Pushing `website` may trigger the configured deployment workflow and therefore requires separate release authorization.

1.3 Likely authorized implementation paths

The smallest anticipated change set is:

Path	Expected responsibility
website/src/pages/cu-time.astro	Expanded governing copy and educational sections.
website/src/components/CUTimeConverter.astro	Additional result fields, details, accessibility, and responsive presentation.
website/src/lib/cuTime/types.ts	Public result contracts for aligned coordinate, age-state, and ratio-state fields.
website/src/lib/cuTime/converter.ts	New ratio derivation and explicit boundary-state semantics.
website/src/lib/cuTime/ui.ts	Formatting and display adapter exposure of already-computed values.

website/src/lib/cuTime/___tests___/converter.test.ts	Core arithmetic and boundary cases.
website/src/lib/cuTime/___tests___/precision-regression.test.ts	Precision and nonfinite-result protection.
website/src/lib/cuTime/___tests___/ui.test.ts	Public display contract.
website/src/lib/cuTime/___tests___/ui-regression.test.ts	Copy, terminology, and layout-regression expectations.

A focused route-content test may be added after inspection of the current test organization.

1.4 Protected implementation paths

Unless a reproducible defect and separate owner decision require otherwise, do not modify:

- website/src/lib/cuTime/constants.ts;
- website/src/lib/cuTime/calendar.ts;
- website/src/lib/cuTime/decimal.ts;
- website/src/lib/cuTime/validation.ts;
- the standalone reference HTML;
- sealed Cosmic Breath structural or content authorities;
- dependency manifests or lockfiles.

2 Part 2 - CU Authority, Statement, and Chronology

2.1 The governing CU Statement

Authority and Source Priority

We are sub z-tomically inclined, countably infinite, composed of foundational elements (the essence of conscious existence), grounded on b-tom (as vast as our shared worlds and their atmospheres), and looking up to c-tom (encompassing the entirety of the cosmos), guided by the uncountable infinite quantum states of intelligence and empowered by God’s Free Will.

For this page, the CU Statement is the governing philosophical and cosmological authority. CU-Time is its mathematical and chronological expression. Empirical science may provide an external observational reference, but it does not create, measure, or govern the full CU framework.

2.2 CU-Time is the full Cosmic Breath coordinate

CU-Time is not a clock beginning at the empirical Big Bang. It belongs to the complete Cosmic Breath:

CU-Time scope

```

sub-ZTOM seed
  -> expansion sequence
  -> ATOM guarded expansion pause
  -> Begin Compression
  -> compression sequence
  -> ZTOM reset boundary
  -> Begin the Next Cosmic Breath
  -> next sub-ZTOM seed

```

The declared model scale is approximately:

$$2.8 \times 10^{12} \text{ years of expansion} + 3.08 \times 10^{11} \text{ years of compression} = 3.108 \times 10^{12} \text{ years per Cosmic Breath.}$$

These are CU model anchors. They are not empirical measurements and must not be recomputed by summing sealed TOM display strings unless separately authorized.

2.3 No zero as a CU state or origin

Authority and Source Priority

Zero is not a state or origin within the Cosmic Universalism chronology. The owner-approved CU framing begins at the sub-ZTOM seed boundary and continues through the ZTOM reset boundary, empowered by God's Free Will. The empirical Big Bang is an internal observable-universe reference within an already-existing CU-Time coordinate.

The implementation may detect equality to an arithmetic reference boundary internally. The public interface must name the boundary rather than presenting that arithmetic condition as a CU state, an origin of existence, or the start of the Cosmic Breath.

The owner-approved one-second seed/reset framing is semantic guidance for the CU-Time page. This feature does not authorize rewriting the sealed Cosmic Breath Explorer's canonical per-state duration strings. Any reconciliation of those separate authorities requires its own owner review.

2.4 Why the Observable Universe appears between B-TOM and C-TOM

Within Cosmic Universalism, the placement of the Observable Universe marker follows the governing CU Statement. The observer is composed of foundational elements, grounded on B-TOM - the scale associated with shared worlds and their atmospheres - and oriented toward C-TOM, the encompassing cosmic scale.

The marker therefore represents the relational standpoint expressed by the CU Statement: conscious existence is grounded within the shared-world scale of B-TOM while looking toward the encompassing cosmic scale of C-TOM. It is a separate, non-selectable CU reference. It is not itself a TOM state, an empirical measurement, or a claim that contemporary science physically locates the Observable Universe between measured layers.

Observer relationship

```
foundational composition
  -> grounded on B-TOM
  -> Observable Universe reference standpoint
  -> looking up to C-TOM
```

3 Part 3 - Current Implementation Inspection

3.1 Current public route

The current website/src/pages/cu-time.astro route contains:

- the page title **A coordinate system for the Cosmic Breath**;
- an **About this converter** notice;
- **Purpose and limits**;
- **Conversion boundaries**;
- the existing CUTimeConverter.astro component.

The current copy correctly identifies CU-Time as a CU mathematical model and states that Gregorian inputs use proleptic Gregorian UTC with explicit CE/BCE civil years.

3.2 Current bidirectional interface

The current component already provides:

1. Gregorian UTC to exact converter-aligned CU-Time;
2. CU-Time plain decimal to Gregorian UTC;
3. fixed-point output with no digits intentionally hidden;
4. six-place scientific notation summaries;
5. copy controls;
6. visible validation and focus movement;
7. a two-column desktop layout that collapses responsively.

These are protected behaviors. The new result data must extend the current cards rather than replacing the converter with a different interaction model.

3.3 Current engine already computes two requested outputs

The inspected TypeScript engine already returns:

Existing engine fields

ForwardConversionResult:

```
jdn
deltaJdn
deltaYears
nasaCuTime
cuTime
yearsSinceBigBang
```

ReverseConversionResult:

```
inputCuTime
nasaCuTime
deltaYears
deltaJdn
jdn
yearsSinceBigBang
Gregorian UTC fields
```

The public adapter currently exposes only CU-Time, scientific notation, and Gregorian UTC. Therefore, the Observable-Universe-Aligned CU Coordinate and the Big Bang reference chronology require public exposure and classification, not new base arithmetic.

3.4 Only the ratio is a new derived quantity

The new ratio is:

$$R = \frac{CU_{OU}}{A_{BB}},$$

where CU_{OU} is the observable-universe-aligned CU coordinate and A_{BB} is the positive elapsed age since the adopted Big Bang reference. The ratio must be produced only when $A_{BB} > 0$.

4 Part 4 - Exact Mathematical Specification**4.1 Approved exact literals**

Internal name	Exact decimal literal
anchorJdn	2451544.5
daysPerYear	365.2421897
baseCuNasa	3094213000000
offset	78955076.49024643522707769749119
nasaUniverseAge	13786999981.453

The internal names remain unchanged for compatibility. Public labels are refined separately. All literals remain strings before Decimal construction. The engine continues to use an isolated Decimal constructor configured for 60 significant digits and half-up rounding.

4.2 Gregorian UTC to aligned coordinate

For a validated Gregorian UTC input, calculate Julian Day Number J with the existing proleptic Gregorian algorithm. Then:

$$\begin{aligned}\Delta J &= J - 2,451,544.5, \\ \Delta Y &= \frac{\Delta J}{365.2421897}, \\ CU_{OU} &= 3,094,213,000,000 + \Delta Y, \\ CU_{full} &= CU_{OU} - 78,955,076.49024643522707769749119, \\ A_{BB} &= 13,786,999,981.453 + \Delta Y.\end{aligned}$$

4.3 CU-Time to Gregorian UTC

For a validated full-cycle CU-Time input:

$$\begin{aligned}CU_{OU} &= CU_{full} + 78,955,076.49024643522707769749119, \\ \Delta Y &= CU_{OU} - 3,094,213,000,000, \\ \Delta J &= \Delta Y \times 365.2421897, \\ J &= 2,451,544.5 + \Delta J.\end{aligned}$$

The current JDN-to-Gregorian algorithm then returns the civil CE/BCE date and UTC time. The same CU_{OU} and A_{BB} values feed the secondary result display.

4.4 Big Bang reference coordinate

The adopted Big Bang reference has a large, nonzero aligned CU coordinate:

$$\begin{aligned}CU_{BB} &= 3,094,213,000,000 - 13,786,999,981.453 \\ &= 3,080,426,000,018.547.\end{aligned}$$

This arithmetic supports the public conceptual rule: the Big Bang is located inside CU-Time and is not coordinate zero for the full Cosmic Breath.

4.5 Ratio calculation

When the entered coordinate is after the adopted Big Bang reference:

$$R = \frac{CU_{OU}}{A_{BB}}.$$

The internal calculation uses Decimal arithmetic. The recommended public summary uses three decimal places

to preserve continuity with the historical reference display, while an expandable detail may expose the full fixed-point value produced by the engine.

4.6 Computational precision and empirical uncertainty

Safety Rule

Sixty-digit Decimal arithmetic preserves the model's literals and reduces floating-point loss. It does not mean the empirical age of the universe is known to sixty digits. Computational precision, model exactness, and empirical measurement uncertainty are different categories.

The page must identify:

- exact model literals and deterministic arithmetic;
- calendar assumptions and supported input resolution;
- the adopted cosmological baseline as an external reference lens;
- the fact that long decimals are model outputs rather than equivalent empirical certainty.

5 Part 5 - Boundary Semantics and Public Output States

5.1 Three age-reference states

The display adapter must classify the Big Bang reference relationship before formatting public text.

Condition	Public age label	Ratio behavior
$A_{BB} > 0$	Derived Age Since the Big Bang	Show the coordinate-to-observable-age ratio.
$A_{BB} = 0$ internally	Big Bang reference boundary	Do not show a numerical ratio; state that the elapsed-age comparison begins at this boundary within already-existing CU-Time.
$A_{BB} < 0$	Pre-Big-Bang Reference Interval: $ A_{BB} $ years	Do not show the ratio because the denominator is not a positive elapsed observable-universe age.

5.2 Approved boundary wording

At the exact adopted boundary, display:

Big Bang reference boundary

The observable-universe elapsed-age comparison begins here within the larger, already-existing CU-Time coordinate.

For the ratio at that boundary, display:

Ratio not shown at the Big Bang reference boundary.

For a coordinate earlier than the adopted reference, display:

Pre-Big-Bang Reference Interval

This coordinate is [absolute interval] years before the adopted observable- universe Big Bang reference within the broader CU-Time coordinate.

Do not display a negative “age” and do not present arithmetic zero as a CU state or origin.

5.3 No conversion failure at the boundary

The primary conversion must still succeed at and before the Big Bang reference. The result continues to show:

- the full CU-Time or Gregorian UTC result;
- the observable-universe-aligned CU coordinate;
- the appropriate boundary-relative chronology;
- a nonnumeric ratio state rather than *Infinity*, *NaN*, or a general conversion error.

6 Part 6 - Public Terminology and Result Architecture

6.1 Primary output: CU-Time - Full Cosmic Breath Coordinate

Recommended definition:

The entered moment expressed within the complete Cosmic Breath coordinate from the sub-ZTOM seed boundary through the ZTOM reset boundary. CU-Time is defined within Cosmic Universalism and does not begin at the empirical Big Bang.

Classification: **CU Mathematical Model**.

6.2 Observable-Universe-Aligned CU Coordinate

Historical internal and source label: NASA CU-Time or *nasaCuTime*.

Recommended definition:

A secondary CU coordinate aligned with the converter’s adopted observable- universe age reference. It places the entered moment within an observable- universe comparison window inside the larger CU-Time coordinate. It is not an official NASA time standard and does not define CU-Time’s origin or duration.

Classification: **CU Mathematical Model using an adopted empirical reference baseline**.

6.3 Derived Age Since the Big Bang

Recommended definition for post-boundary dates:

The elapsed observable-universe age derived from the converter's adopted cosmological baseline and the entered coordinate. It is not elapsed time since sub-ZTOM and is not the age of the complete Cosmic Breath.

Classification: **Model-derived observable-universe reference.**

6.4 CU Coordinate-to-Observable-Age Ratio

Recommended definition:

A dimensionless CU mathematical comparison between the observable-universe- aligned CU coordinate and the positive derived elapsed age since the adopted Big Bang reference. It is not a standard NASA, Planck, or contemporary cosmology statistic and does not measure the physical size of the Big Bang.

Classification: **CU Mathematical Comparison.**

6.5 Historical terminology disclosure

The page may state once, in expandable method content:

The standalone historical converter calls the secondary coordinate "NASA CU-Time." The website uses "Observable-Universe-Aligned CU Coordinate" to avoid implying that NASA created or endorses the CU scale. The internal `nasaCuTime` identifier remains unchanged for compatibility.

7 Part 7 - Interface and Content Design

7.1 One converter, automatic secondary results

Both directions retain their current forms and primary actions. No additional input is required because all three secondary values derive from the same validated conversion result.

Forward result hierarchy

Conversion complete

Input instant

CU-Time - Full Cosmic Breath Coordinate

Scientific notation

Observable-universe reference

Observable-Universe-Aligned CU Coordinate

Derived Age Since the Big Bang / boundary-relative state

CU Coordinate-to-Observable-Age Ratio / nonnumeric state

Reverse result hierarchy

Conversion complete

Input CU-Time

Scientific notation

Corresponding Gregorian UTC

Observable-universe reference

Observable-Universe-Aligned CU Coordinate

Derived Age Since the Big Bang / boundary-relative state

CU Coordinate-to-Observable-Age Ratio / nonnumeric state

7.2 Result-card presentation

The primary result remains visually dominant. The secondary group should:

- appear automatically after successful conversion;
- use a visible **Observable-Universe Reference** group heading;
- preserve long-number wrapping without horizontal overflow;
- provide a concise value plus an optional expandable full-precision and method explanation;
- avoid repeating the full educational essay inside each card;
- reset disclosure state when a new conversion replaces the result.

7.3 Copy behavior

The current primary copy controls remain. A future implementation may add individual copy controls for exact aligned coordinate and exact age only if the UI remains uncluttered and tests cover accessible names and status messages. The ratio summary should not replace the primary copy action.

7.4 Expanded educational sections below the converter

Recommended page sections:

1. **CU-Time and the Governing CU Statement;**
2. **The Full Cosmic Breath Coordinate;**
3. **The Observable-Universe Reference Window;**
4. **Why the Big Bang Does Not Begin CU-Time;**
5. **Why the Observable Universe Appears Between B-TOM and C-TOM;**
6. **How Gregorian UTC Becomes Julian Day Number;**
7. **How the Aligned Coordinate and Offset Work;**

8. **How the Big Bang Reference Chronology Is Derived;**
9. **How the Ratio Works;**
10. **Computational Precision and Empirical Uncertainty;**
11. **Conversion Boundaries and Historical Terminology.**

The sections must distinguish CU propositions, model arithmetic, empirical reference, and historical wording rather than mixing them in one undifferentiated paragraph.

8 Part 8 - TypeScript Data and Formatting Design

8.1 Recommended internal model

The engine already returns `nasaCuTime` and `yearsSinceBigBang`. Add a small derived-state contract rather than renaming existing fields.

Recommended conceptual types

```
type ObservableAgeState =
  | { kind: 'elapsed'; years: Decimal }
  | { kind: 'big-bang-boundary' }
  | { kind: 'pre-big-bang'; intervalYears: Decimal };

type ObservableRatioState =
  | { kind: 'available'; value: Decimal }
  | { kind: 'not-applicable'; reason: 'boundary' | 'pre-big-bang' };
```

Exact names may change during implementation review, but the discriminated state behavior is required.

8.2 Recommended public display contract

Forward display result

```
GregorianToCuTimeDisplayResult {
  gregorianUtc
  cuTime
  cuTimeExponential
  observableUniverseAlignedCuTime
  observableUniverseAlignedCuTimeExponential
  observableAge
  coordinateToObservableAgeRatio
}
```

Reverse display result

```
CuTimeDisplayResult {
  inputCuTime
}
```

```
inputCuTimeExponential
gregorianUtc
observableUniverseAlignedCuTime
observableUniverseAlignedCuTimeExponential
observableAge
coordinateToObservableAgeRatio
}
```

Boundary-aware objects are preferable to magic strings inside the arithmetic engine. Human-readable labels belong in the UI adapter or presentation layer.

8.3 Formatting policy

Value	Recommended display
Full CU-Time	Exact fixed-point value plus six-place scientific notation, preserving current behavior.
Observable-universe-aligned coordinate	Exact fixed-point value plus optional six-place scientific notation.
Elapsed age	Exact or appropriately grouped fixed-point model value; clearly label model precision.
Ratio	Three-decimal summary; optional expandable full fixed-point value.
Boundary/pre-boundary	Named semantic state; no numerical zero-age or negative-age label.

Do not use the historical readable magnitude formatter that may double-count a fractional remainder. Use the current TypeScript Decimal values directly.

9 Part 9 - Accessibility, Responsive Design, and Interaction

9.1 Accessibility requirements

The enhancement must preserve and extend:

- semantic forms, fieldsets, legends, labels, and error associations;
- aria-live result and copy status behavior;
- focus movement to the result after a successful conversion;
- focus return to the invalid field after validation failure;
- semantic dl, dt, and dd result structures;
- keyboard-operable details/disclosure controls;
- visible focus for every new interactive control;
- accessible names that distinguish primary and secondary copy actions;
- no meaning conveyed only through color.

9.2 Responsive requirements

Verify at approximately 1440, 1024, 768, 390, and 320 pixels. The expanded result cards must avoid horizontal overflow and keep long Decimal values selectable and readable. At narrow widths, the two converter directions remain stacked and the secondary result group must not force fixed-width columns.

9.3 Progressive disclosure

Use native details/summary or the established disclosure pattern for:

- full-precision secondary values when a concise summary is shown;
- “How this is calculated” explanations;
- historical source terminology;
- computational precision versus empirical uncertainty.

Do not hide the existence or classification of the three secondary outputs; only their extended method text may be collapsed.

10 Part 10 - Test Plan and Acceptance Criteria

10.1 Core engine tests

Add or update tests for:

- forward and reverse aligned-coordinate parity;
- forward and reverse derived-age parity;
- ratio arithmetic using Decimal values;
- ratio state when elapsed age is positive;
- Big Bang reference boundary state without division by zero;
- pre-Big-Bang interval state without negative-age wording;
- no Infinity, NaN, or nonfinite public result;
- unchanged anchor, calendar, CE/BCE, validation, and round-trip behavior.

10.2 UI adapter tests

Verify both conversion directions return:

- the unchanged primary result fields;
- exact aligned coordinate and scientific notation;
- a discriminated observable-age display state;

- a discriminated ratio display state;
- canonical labels and historical terminology disclosure;
- no external-converter URLs.

10.3 Component regression tests

Verify:

- the three secondary result labels exist in both cards;
- all new data hooks are unique and correctly bound;
- copy controls retain accessible status regions;
- result focus remains intact;
- details controls have correct accessible text;
- the route retains its CU Mathematical Model classification;
- the page states that CU-Time does not begin at the Big Bang;
- the page states why the Observable Universe marker is relationally between B-TOM and C-TOM;
- prohibited external converter links are absent.

10.4 Reference vectors

At the arithmetic anchor 01/01/2000 CE 00:00:00 UTC:

Field	Expected value
JDN	2451544.5
Delta years	0
Observable-universe-aligned CU coordinate	3094213000000
Full CU-Time	3094134044923.50975356477292230250881
Derived elapsed age	13786999981.453

For 07/23/2026 CE 00:00:00 UTC, the 60-digit engine produces:

July 23, 2026 reference vector

Observable-Universe-Aligned CU Coordinate:
3094213000026.55772053049872513126048647167006073833096395983

Full CU-Time:
3094134044950.06747409527164743376929647167006073833096395983

Derived Age Since the Big Bang:
13787000008.0107205304987251312604864716700607383309639598297

Ratio summary:
224.430

The reference vector must be confirmed by the live TypeScript engine before it is committed as a test expectation.

10.5 Full acceptance gate

A release candidate must pass:

CU-Time expanded-output verification gate

```
cd /Users/cosmos/Documents/GitHub/CosmicUniversalismStatement/website || exit 1

npm test
npm run build

cd ..
git diff --check
git status --short --branch
git diff --stat
```

Manual browser acceptance must cover both directions, CE/BCE, boundary states, keyboard navigation, copy behavior, desktop/tablet/mobile layouts, and a clean browser console.

11 Part 11 - Controlled Implementation Phases

11.1 Phase 1 - Pure result semantics

Authorized objective:

- add the ratio derivation;
- add elapsed/boundary/pre-boundary state semantics;
- preserve constants and calendar behavior;
- add focused core tests.

Completion gate: pure tests pass with no component or route edits.

11.2 Phase 2 - Public display adapter

Authorized objective:

- expose the existing aligned coordinate and age results;
- format the three semantic age states;
- format the ratio availability state;

- preserve all current display fields and validation outcomes.

Completion gate: UI adapter tests pass and no HTML is yet changed.

11.3 Phase 3 - Converter result cards

Authorized objective:

- add the secondary result group to both directions;
- preserve primary visual hierarchy;
- add method disclosures and classifications;
- preserve focus, copy, keyboard, and responsive behavior.

Completion gate: focused component tests and manual browser inspection pass.

11.4 Phase 4 - Route education

Authorized objective:

- add CU Statement governance;
- explain full-Cosmic-Breath CU-Time;
- explain the Big Bang as an internal reference boundary;
- explain B-TOM to C-TOM observer placement;
- explain formulas, precision, empirical uncertainty, and historical terminology;
- exclude external converter links.

Completion gate: route-content tests, accessibility review, and source-boundary review pass.

11.5 Phase 5 - Full acceptance

Run the complete test/build gate, inspect the full diff, verify no protected file changed, and perform manual browser acceptance. Do not commit or push without separate owner authorization.

12 Part 12 - Copy-and-Paste Implementation Prompts

12.1 Feature-branch preflight prompt

Copy-and-Paste Prompt: CU-Time Expanded Outputs Preflight

```
Repository:
/Users/cosmos/Documents/GitHub/CosmicUniversalismStatement

Protected branch:
```

```
main

Stable branch:
website

Expected website baseline:
e1c82bff1d37732f52b03fa4734cf7e4120b71e9

Proposed feature branch:
feature/cu-time-expanded-outputs

Before any edit:

1. Verify origin, branch, HEAD, and worktree.
2. Confirm website is at the expected baseline.
3. Confirm the feature branch does not already exist locally or remotely.
4. Inspect only the authorized CU-Time paths.
5. Report the smallest phase-one file set and tests.

Do not edit, install dependencies, commit, push, merge, tag, release,
or deploy.
```

12.2 Phase-one implementation prompt

Copy-and-Paste Prompt: Implement CU-Time Result Semantics Only

Implement only the pure CU-Time result semantics approved by the CU-Time Expanded Outputs and Method Implementation Manual v1.1.

Required behavior:

- preserve all exact constants;
- preserve Gregorian/JDN algorithms;
- preserve current forward and reverse primary results;
- calculate coordinate-to-observable-age ratio only when the derived elapsed age is positive;
- represent the exact Big Bang arithmetic boundary as a named boundary state, not a public zero-age CU state;
- represent earlier coordinates as a positive pre-Big-Bang reference interval, not a negative age;
- do not emit Infinity or NaN;
- do not edit the Astro route or component in this phase.

Add focused tests.

Do not commit, push, merge, tag, release, or deploy.

12.3 Final review prompt

Copy-and-Paste Prompt: Final CU-Time Expanded Outputs Review

Perform a final no-edit release-candidate review of the CU-Time expanded outputs feature.

Verify:

1. repository branch, HEAD, and worktree;
2. exact constants and Decimal precision unchanged;
3. Gregorian and reverse calendar behavior unchanged;
4. aligned coordinate and age values match in both directions;
5. ratio behavior for positive elapsed age;
6. named Big Bang boundary behavior;
7. pre-Big-Bang interval behavior;
8. no Infinity, NaN, negative-age label, or zero-as-CU-origin copy;
9. primary result hierarchy preserved;
10. both result cards expose all approved secondary values;
11. CU Statement governance and full-Cosmic-Breath interpretation;
12. B-TOM/C-TOM Observable Universe explanation;
13. no external converter links;
14. accessibility, keyboard, focus, copy, and responsive behavior;
15. focused and full test results;
16. Astro build and eight-route output;
17. no protected or unrelated file changes;
18. deployment implications of any later website push.

Do not edit, commit, push, merge, tag, release, or deploy.

13 Part 13 - Commit, Merge, and Deployment Boundaries

13.1 Commit boundary

After full acceptance, an explicit owner authorization may permit staging only the verified CU-Time feature files. Use exact paths and inspect the staged diff. The commit must not include the reference HTML, sealed Cosmic Breath ledgers, manual source files unless separately approved, or unrelated documentation.

13.2 Merge boundary

A future merge should use an explicit non-fast-forward merge into `website` only after post-feature verification. Because pushes to `website` may publish GitHub Pages, merge/push authorization must be clear about deployment consequences.

13.3 Tag boundary

Do not move `website-v1.1-cosmic-breath`. A CU-Time expanded-output tag, GitHub Release, or deployment record requires a new owner-approved name and a separate action.

14 Appendix A - Source Authority Record

Source	Role	Use in this manual
Owner decisions in this workflow	Governing authority	Product scope, terminology, no-zero CU ontology, Big Bang boundary semantics, B-TOM/C-TOM explanation.
CU Statement	Canonical CU framing	Governs CU-Time meaning and observer relationship.
Live CU-Time TypeScript engine	Production arithmetic	Establishes current constants, fields, calendar path, and already-computed aligned coordinate/age.
Website Master Manual v1.1	Governance and house style	Current branch/release discipline, classifications, testing, and LaTeX standard.
Cosmic Breath Time Converter Manual v1.0	Historical implementation authority	Preserves exact-literal, parity, module, test, and unchanged-reference rules.
Standalone converter HTML v3.0.1	Behavioral and provenance reference	Establishes historical labels and output relationships; remains unchanged.
Time Calculation and CU Framework files	Historical/open CU research	May inform context but do not silently override sealed authorities, owner decisions, or production arithmetic.

15 Appendix B - Owner Decision Record

ID	Approved decision
CTE-OD-01	Preserve one bidirectional converter; calculate all secondary outputs automatically.
CTE-OD-02	Add Observable-Universe-Aligned CU Coordinate, Derived Age Since the Big Bang, and CU Coordinate-to-Observable-Age Ratio.
CTE-OD-03	Exclude links and calls to action for other converter pages.
CTE-OD-04	CU-Time is the full sub-ZTOM-to-ZTOM Cosmic Breath coordinate and does not begin at the Big Bang.
CTE-OD-05	The CU Statement governs the page; empirical science is a secondary observational lens.
CTE-OD-06	The Observable Universe marker is relationally between B-TOM and C-TOM because the observer is composed of foundational elements, grounded on B-TOM, and looking up to C-TOM.
CTE-OD-07	Zero is not a CU state or origin; the public page names the Big Bang reference boundary rather than displaying a zero-age CU state.
CTE-OD-08	Before the adopted Big Bang reference, display a positive pre-Big-Bang reference interval rather than a negative age.
CTE-OD-09	Do not show a ratio at or before the adopted Big Bang reference boundary.

CTE-OD-10

Preserve the original reference HTML and exact constants; do not silently alter converter mathematics.

16 Appendix C - Final Acceptance Checklist

- ☐ Correct feature branch and clean starting baseline.
- ☐ Exact constants and Decimal precision unchanged.
- ☐ Existing forward and reverse conversions preserved.
- ☐ Aligned coordinate exposed in both directions.
- ☐ Big Bang chronology exposed with three semantic states.
- ☐ Ratio available only for positive elapsed age.
- ☐ No numerical zero-age CU state, negative-age label, Infinity, or NaN.
- ☐ No external converter links.
- ☐ CU Statement and full Cosmic Breath explanation present.
- ☐ B-TOM/C-TOM Observable Universe explanation present.
- ☐ Computational precision separated from empirical uncertainty.
- ☐ Accessibility and responsive acceptance complete.
- ☐ Focused tests and full suite pass.
- ☐ Astro production build passes with expected routes.
- ☐ No unrelated or protected files changed.
- ☐ No commit, push, merge, tag, release, or deployment without separate authorization.

17 Appendix D - Manual Revision Record

Version	Date	Revision
1.0	July 19, 2026	Historical Cosmic Breath Time Converter migration manual establishing parity, exact constants, TypeScript extraction, validation, tests, and Astro integration.
1.1	July 23, 2026	Adds the owner-approved expanded-output specification: one automatic bidirectional result model, observable-universe-aligned coordinate, Big Bang boundary-relative chronology, ratio semantics, CU Statement governance, full-Cosmic-Breath interpretation, B-TOM/C-TOM observer placement, updated implementation phases, and acceptance tests.